

TrES-3b

R-0502

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240703 · created 2026-07-20T16:59:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460494.7929 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.057
Transit depth (Rp/Rs)^2	3.32 +/- 2.08 [%]
Orbital Inclination (inc)	81.85 +/- 2.94
Airmass coefficient 1 (a1)	16.45 +/- 11.1
Airmass coefficient 2 (a2)	-2.53 +/- 1.51
Scatter in the residuals of the lightcurve fit is	70.38 %
Best Comparison Star	#1 - [348, 68]
Optimal Aperture	2.06
Optimal Annulus	5.51
Transit Duration (day)	0.049 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.057 vs 0.16309 (deviation 0.31σ)
Duration fit vs published	0.049 d vs 0.0567 d (deviation 0.25σ)
Mid-time O–C	2.1 min
Depth SNR	3.0
Residual scatter	70.38 %

Light curve

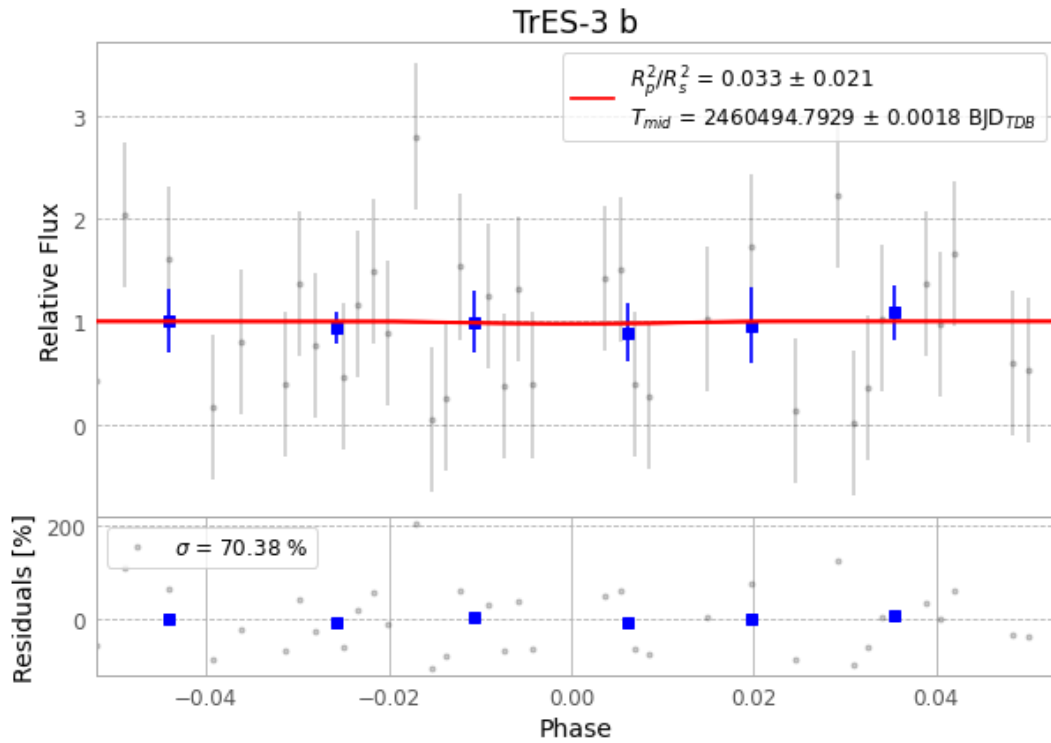


Figure 1 — FinalLightCurve_TrES-3 b_2024-07-02.png (SHA-256 5ac0080a5d7f2b97...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [615, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 231ac81da7acdcdc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

68 FITS frames (45 MB), TRES-3240703051815.FITS → TRES-3240703083915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240703.log (2c6bff1701e0...), FinalLightCurve_TrES-3 b_2024-07-02.png (5ac0080a5d7f...), FinalLightCurve_TrES-3 b_2024-07-02.csv (6bf2165cea3c...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 16:59Z.